

Functional Requirements Specification (FRS)

Loan Origination System (LOS) – Gold Loan

1. Purpose

This FRS defines the functional requirements for a Loan Origination System (LOS) covering user authentication, role-based access, customer management, rule-engine configuration, loan-product configuration, application processing, approval controls, auditability and the Gold Loan workflow described in the provided source document.

The provided source specifies a Gold Loan LOS covering field inventory, modules, document/API requirements and an end-to-end flow from lead/walk-in capture through closure and collateral release. It states that default and auction-related collections/recovery workflow is out of scope for LOS at this phase.

2. User Roles and Permissions

Role	Access Level	Permissions
Super Admin	Full administrative access	Create/modify user roles; make users Maker/Checker/Viewer; add new customer; fill/edit loan forms; create/modify products and rules; configure branches/vaults/templates/rates/commission settings; view all records; access audit logs.
Maker	Loan/application data entry	Create and update applications assigned/available to Maker; capture customer/KYC, appraisal, loan and documentation information; submit completed application to Checker; view own/assigned records. Cannot perform final checking/approval of own submission.
Checker	Verification/checking	View Maker-submitted applications; verify data/documents/rules; approve, reject, return for correction or place on hold as configured; record checker remarks; cannot alter Maker-submitted data without controlled return/reopen workflow.
Viewer	Read-only	View permitted dashboards, customers, applications, products/rules and reports. No create, edit, submit, approve, reject or configuration permission.

Maker–Checker control: a Maker must not be able to complete the Checker action on the same transaction. The system must retain the actor, timestamp and action history.

3. Role Permission Matrix

Function	Super Admin	Maker	Checker	Viewer
Login / OTP	Yes	Yes	Yes	Yes
Dashboard	Yes	Yes	Yes	Yes
Add Customer	Yes	Yes	No	No
Edit Customer	Yes	Yes	No	No
Create Loan Application	Yes	Yes	No	No
Fill / Edit Loan Form	Yes	Yes	No	No
Submit to Checker	Yes	Yes	Yes	No
Check Application	Yes	Yes	Yes	No
Approve / Reject / Return/Comments/Edit Application	Yes	Yes	Yes but only for the comments else No	No
Create Rule Engine	Yes	No	No	No
Create / Configure Product	Yes	yes	No	No
Configure Branch / Vault / Templates	Yes	No	No	No
View Reports / Audit Trail	Yes	Yes	Yes	Yes

4. Page-wise Functional Requirements

4.1 Login

Sr. No.	Field Name	M/O	Character length	String type	Description
FR-LOGIN-001	Login user ID	M	8–50	Alphanumeric	System shall provide Login ID input.
FR-LOGIN-002	Password	M	8–20	Alphanumeric (Masked) Jwt Token for encryption	System shall provide password input with masked characters.

FR-LOGIN-003	Credential Validation	M	N/A	System Validation	System shall validate credentials and active user status.
FR-LOGIN-004	Two-Step Verification	M	6	Numeric (OTP)	After valid credentials, system shall send/validate OTP for second-factor verification.
FR-LOGIN-005	OTP Controls	M	N/A	System Configuration	System shall support OTP expiry, resend and configurable attempt limits.
FR-LOGIN-006	Role Loading	M	N/A	System Function	After successful OTP, system shall load permissions associated with the user's role.
FR-LOGIN-007	Failed Login	M	N/A	System Function	System shall record failed attempts and apply configurable lockout/security controls.

For the **LOS Login – Password** requirement, you can define comprehensive password restrictions like this:

– Password Requirements are below

The system shall provide a Password field where the user must enter a password that complies with the following restrictions:

1. Password shall contain a **minimum of 8 characters**.
2. Password shall contain a **maximum of 16 characters**.
3. Password shall contain at least **one uppercase letter (A–Z)**.
4. Password shall contain at least **one lowercase letter (a–z)**.
5. Password shall contain at least **one numeric digit (0–9)**.

6. Password shall contain at least **one special character**, such as @, #, \$, %, &, *, !.
7. Password shall **not contain spaces**.
8. Password shall **not contain the user's Login User ID**.
9. Password shall **not contain the user's registered mobile number**.
10. Password shall **not contain commonly used passwords** such as Password@123, Admin@123, or 12345678.
11. Password shall **not be the same as any of the user's previous 5 passwords**.
12. Password shall be **case-sensitive**.
13. The system shall not display the password in plain text while entering it.
14. The system shall provide a **Show/Hide Password** option.

4.2 Forget Password

Sr. No.	Field Name	M/O	Character length	String type	Description
FR-FORGO T-001	Forgot Password	M	—	Link/Button	A Forgot Password? option shall be displayed below the Login button on the LOS Login page.
FR-FORGO T-002	Forgot Password Navigation	M	—	Link	On clicking Forgot Password?, the user shall be redirected to the Forgot Password page.
FR-FORGO T-003	Registered Email ID	M	100	Email	The user shall enter the registered email ID associated with the LOS account.
FR-FORGO T-004	Email Format Validation	M	100	Validation	The system shall validate that the entered email ID is in a valid email format.

FR-FORGO T-005	Email ID Verification	M	100	Validation	The system shall verify whether the entered email ID is registered with an active LOS user account.
FR-FORGO T-006	Send Reset Link	M	—	Button	On successful email validation, the system shall send a password reset link to the registered email ID.
FR-FORGO T-007	Reset Password Link	M	—	Link	The user shall receive a password reset link on the registered email ID.
FR-FORGO T-008	Reset Password Page	M	—	Page	On clicking the reset password link, the user shall be redirected to the Reset Password page.
FR-FORGO T-009	Old Password	M	8–20	Password	The user shall enter the existing/old password for verification.
FR-FORGO T-010	New Password	M	8–20	Password	The user shall enter a new password that satisfies the configured password rules.
FR-FORGO T-011	Confirm New Password	M	8–20	Password	The user shall re-enter the new password for confirmation.
FR-FORGO T-012	OTP	M	6	Numeric	An OTP shall be sent to the registered

					email ID and an OTP field shall be available on the same Reset Password page.
FR-FORGO T-013	OTP Verification	M	6	Validation	The system shall verify that the OTP entered by the user matches the OTP sent to the registered email ID.
FR-FORGO T-014	OTP Expiry	M	—	Validation	The OTP shall be valid only for the configured time period. An expired OTP shall not be accepted.
FR-FORGO T-015	Resend OTP	O	6	Button	The user shall be provided with a Resend OTP option if the OTP is not received or has expired.
FR-FORGO T-016	Old Password Validation	M	8–20	Validation	The system shall verify that the entered old password is correct.
FR-FORGO T-017	New Password Validation	M	8–20	Validation	The system shall validate the new password against the configured password complexity rules.
FR-FORGO T-018	Old and New Password Check	M	8–20	Validation	The system shall not allow the user to set the new password same as the

					old password.
FR-FORGO T-019	Password Match Validation	M	8–20	Validation	The system shall verify that New Password and Confirm New Password are identical.
FR-FORGO T-020	Password Match Validation	M	8–20	Validation	The system shall verify that New Password and Confirm New Password are identical.
FR-FORGO T-021	Reset Password Button	M	—	Button	The user shall click Reset Password after entering all required details and a valid OTP.
FR-FORGO T-022	Password Reset Success	M	—	Message	After successful validation, the system shall update the user's password and display a successful password reset message.
FR-FORGO T-023	Password Reset Failure	M	—	Message	If any validation fails, the system shall display an appropriate error message and shall not update the password.
FR-FORGO T-024	Login with New Password	M	—	Navigation	After successfully resetting the password, the user shall be able to log in using the new password.

FR-FORGO T-025	Reset Link Expiry	M	—	Validation	The password reset link shall expire after the configured validity period.
FR-FORGO T-026	Invalid Reset Link	M	—	Validation	If the reset link is invalid, expired, or already used, the system shall display an appropriate error message.
FR-FORGO T-027	OTP Retry Limit	M	—	Validation	The system shall restrict the number of incorrect OTP attempts according to the configured security policy.
FR-FORGO T-028	Password Reset Audit	M	—	Audit Log	The system shall maintain an audit record of successful and failed password reset attempts.

4.2 Dashboard

ID	Function	M/O	Requirement
FR-DASH-001	Application Overview	M	Dashboard shall show application counts by status.
FR-DASH-002	Role-based View	M	Dashboard widgets/data shall be restricted according to user role.
FR-DASH-003	Maker Queue	M	Maker shall see draft, assigned and returned applications relevant to the Maker.
FR-DASH-004	Checker Queue	M	Checker shall see applications awaiting checking.
FR-DASH-005	Admin Overview	M	Super Admin shall see overall application/customer/user/product/rule summaries.
FR-DASH-006	Search/Filter	M	Dashboard shall support configurable filters such as date, branch, product and status.

FR-DASH-007	Alerts	O	System may show pending actions, exceptions, SLA alerts and failed integrations.
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4.3 Customer

ID	Function	M/O	Requirement
FR-CUST-001	Create Customer	M	Authorized users shall create a customer/entity record.
FR-CUST-002	Customer Type	M	Capture customer type/constitution as applicable to the product.
FR-CUST-003	Duplicate Check	M	System shall check duplicate customer/application identifiers before creation.
FR-CUST-004	KYC Data	M	Capture applicable KYC information and verification status.
FR-CUST-005	Contact Details	M	Capture mobile/email/address required for communication and verification.
FR-CUST-006	Ownership Declaration	M	For Gold Loan, capture ownership declaration of pledged gold.
FR-CUST-007	End-use Declaration	M	For Gold Loan, capture end-use declaration; the provided BRD states purchase of gold cannot be declared as the purpose.
FR-CUST-008	KYC Integration	M	Where configured, invoke PAN/Aadhaar/DigiLocker/face-match services and store verification result/reference.
FR-CUST-009	Customer Search	M	Authorized users shall search and view customer profiles.
FR-CUST-010	Customer Audit	M	Customer creation and material updates shall be audit logged.